

Alexander Brown

San Jose, CA | 408.355.5966 (cell) | alexjirocoll@gmail.com

EDUCATION

Georgia Institute of Technology

Online M.S. Computer Science

August 2025 - *Expected May 2027*

University of California, Irvine

B.S. Computer Science

September 2022 - *June 2025*

Intelligent Systems Specialization

- GPA: 3.85/4.0

PROJECTS & ACTIVITIES

AI BWC Footage Tagging and Purging

Signal Hill, CA

Capstone Project Team Lead

January 2025 - June 2025

- 1st Place Computer Science Winner at UCI ICS Expo
- Collaborated with the Signal Hill Police Department to develop AI-driven computer vision software for identifying and categorizing police body-worn camera footage
- Implemented ensemble machine learning models to classify 16 terabytes of untagged footage, incorporating Imagebind, Efficientnet, and transformer models

Reinforcement Learning Agent: Cuphead

Irvine, CA

AI Project

January 2025 - March 2025

- Built a reinforcement learning agent to defeat Cuphead's 'Root Pack' boss by combining YOLO-based real-time object detection with a Deep Q-Network (DQN), demonstrating RL in fast-paced, visually complex environments

RESEARCH

Machine Learning Research Assistant

Cal IT2, CA

California Institute of Telecommunications & Information Technology

Sept 2024 - April 2025

- Collaborated with a postdoctoral researcher in thermal energy systems
- Developed predictive supervised machine learning models using PyTorch that leverage time series data to simulate thermal energy storage system behavior, significantly reducing computational overhead compared to traditional simulation methods

WORK EXPERIENCE

Full-Stack Web Developer

University of California, Irvine

Student Affairs, Information Technology

March 2023 - June 2025

- Developed and maintained full-stack web applications (Laravel, PHP, JavaScript, CSS) for 9,000+ annual users; automated workflows with Cypress testing and scripts

RELEVANT COURSEWORK

Databases, Systems, Machine Learning, Artificial Intelligence, Data Structures, Algorithm Analysis, Information Retrieval, Embedded Systems, Graph Algorithms, Probability, Statistics

SKILLS

Languages & Tools: Python, SQL, Java, C++, Git, Bash, Docker, AWS

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Laravel, MySQL, pandas, NumPy

Techniques: Machine Learning, Data Analysis, Backend Testing, CI/CD, End-to-End Testing